Which Factors Contribute to the Accuracy and Efficiency of Horse Racing Betting Markets?

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ABSTRACT

This is the abstract.

- A brief introduction to the objectives.
- A brief description of the main work.
- A brief description of the contributions, major findings, results achieved and conclusion.

Nomenclature

BSP — Betfair Starting Price

1. Introduction

1.1. Project Aims & Objectives

The primary aim of this project is to determine which factors are most significant in influencing the accuracy and efficiency of horse racing markets. There are many different ways to classify horse races including but not limited to time, distance and race type (Hurdles, Chase, Handicap, etc.). Additionally, races can be classified by their market factors such as trade volume or overround.

Accuracy of the market is a measure of how well the listed odds represent the true odds of a horse winning. In a market that was 100% accurate, the odds given would provide a 0% return on all winning bets. To determine the accuracy of the odds, I will be comparing various test statistics such as the Brier score and Log-Loss.

An efficient market is one in which prices are said to ''fully reflect'' all of the available information and are shown to be *weak-form* efficient when this can be demonstrated using an information set made exclusively of historical price data (Fama, 1970). A systematic betting strategy using historical pricing data that is able to generate a consistent profit demonstrates a lack of *weak-form* efficiency within a market (Hausch, 1981). In order to determine which groups of races are *weak-form* efficient, I shall apply various betting strategy profiles to them in isolation. Some of these strategies will be simple and some will be adapted from previous work.

1.2. Background & Literature survey

1.2.1. Literature Survey

1.2.2. Betfair & Bookmaking

There are two typical forms of betting markets available for horse racing generally: fixed-price bookmakers and tote operators.

A fixed-price bookmaker will quote prices (odds) at which they are willing to sell bets at. If the bet pays out, the bettor receives the return originally quoted to them. The bettor is given a guarantee of their winnings before placing the bet. It is important to note that it is unlikely a bookmaker will allow a bettor to ‘‘lay’’ a bet, that is to bet that something *will not* happen. Bookmakers make their money by appropriately setting odds such that in all cases (or at least over a period of time), they will ensure a profit. This is done by changing odds to account for the amount of money bet already (balancing the books) and by providing unfair odds that do not sum to one.

A tote (otherwise known as parimutuel betting) is a system in which money for all bets is placed into a pool and then distributed to winning bettors. This means that bettors are not given an exact payout quote at the time of purchasing their bet, but they may be given an estimate. Tote operators make money by taking a certain percentage of the pool for themselves, typically called the ‘‘track take’’.

Betfair is a UK based company which operates a betting exchange service. A betting exchange is different from a typical bookmaker in that instead of the operator quoting odds prices or offering a tote service, it allows users to form bets with one another.

2. Background Theories

3. Main Results

3.1. Objectives, Specification & Design

3.2. Methodology & Implementation

3.2.1. Data Acquisition

3.2.2. Software Development

3.3. Results, Analysis & Evaluation

3.3.1. Accuracy

3.3.2. Efficiency

4. Legal, Social, Ethical & Professional Issues

- Use of live animals?

- Profit Derived from gambling losers
- Similarity to stock market
- legality of companies betting

5. Conclusion

References

Fama, Eugene F., “Efficient capital markets: a review of theory and empirical work,” *The Journal of Finance* **25**(2), pp. 383-417, [American Finance Association, Wiley] (1970).

Hausch, Donald B.,Ziemba, William T., Rubinstein, Mark, “Efficiency of the market for race-track betting,” *Management Science (pre-1986)* **27**(12), p. 1435, Institute for Operations Research and the Management Sciences, Linthicum, United States (1981-12).

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